

AI Impact on Student Performance

Executive Dashboard — An Analysis of AI Tool Usage vs Academic Outcomes across 8,000 Students

What This Project Covers

A 4-page interactive Excel dashboard exploring whether AI tool usage (ChatGPT, Gemini, Claude, Copilot) helps or hurts student academic performance, built from a dataset of 8,000 student records.

1

Objective

Measure the relationship between AI adoption, AI dependency, and final academic scores.

2

Dataset

8,000 student records covering grade level, gender, study habits, AI usage, and exam results.

3

Dashboard Pages

Overview, Performance, Study Habits, and Demographics — each with KPIs and interactive charts.

4

Key Question

Does using more AI tools/content correlate with higher or lower final scores?

Data Cleaning Steps

Steps taken in Excel to prepare the raw student dataset before analysis and dashboarding.

1

Removed Duplicates & Blank Rows

Used Remove Duplicates and filtered blank Student IDs to ensure one clean record per student.

2

Standardized Categorical Values

Unified inconsistent entries (e.g. "chatgpt" / "ChatGPT" / "CHAT GPT") into single categories such as ChatGPT, Gemini, Claude, Copilot, None.

3

Handled Missing Values

Filled or flagged missing Study Hours, Sleep Hours, and AI Usage fields using averages or "Not Reported" labels.

4

Fixed Data Types & Formats

Converted percentages, scores, and hours to consistent numeric formats; corrected text stored as numbers.

5

Outlier Review

Checked Study Hours, Sleep Hours, and Final Scores for unrealistic values and corrected or removed them.

6

Created Helper Columns

Added Performance Category (High/Medium/Low), AI User flag, and Grade Level grouping for pivoting and slicers.

Data Analysis Method

How the cleaned dataset was turned into KPIs, comparisons, and correlation insights.

1	Pivot Tables	Built pivot tables to summarize scores by grade level, gender, AI tool, and performance category.
2	KPI Cards	Calculated headline metrics — pass rate, AI adoption rate, average scores — using AVERAGEIFS / COUNTIFS.
3	Correlation Analysis	Used CORREL() to test the relationship between AI dependency, AI-generated content %, and final score.
4	Segmentation	Grouped students by AI usage purpose, dependency level, and study/sleep hours to compare outcomes.
5	Trend & Distribution Charts	Applied line, scatter and stacked charts to visualize score trends and performance distribution.
6	Interactive Filtering	Connected slicers (Grade Level, Gender, Performance Category, AI Users) across all 4 dashboard pages.

Overview — Executive Dashboard

Full-page view: adoption rate, score gaps, and correlation snapshot across all 8,000 students.

Total Students

8,000

AI Adoption Rate

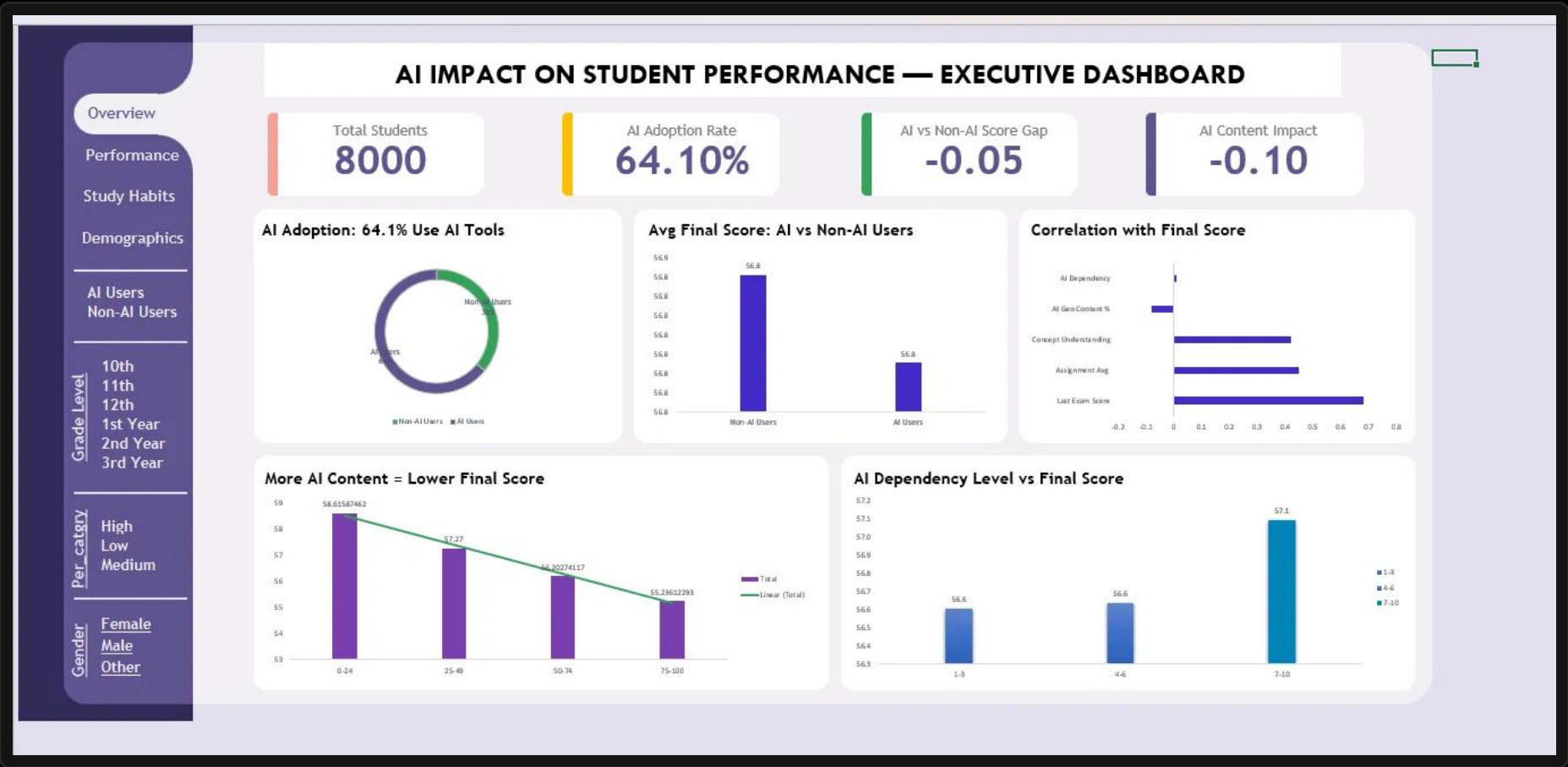
64.10%

AI vs Non-AI Score Gap

-0.05

AI Content Impact

-0.10



Overview — Chart Explanations

Chart-by-chart breakdown of what each visual shows and why it matters.

1

AI Adoption: 64.1% Use AI Tools (Donut Chart)

Splits the student base into AI Users and Non-AI Users. Nearly two-thirds of students report using at least one AI tool, confirming AI adoption is now mainstream rather than a niche behavior.

2

Avg Final Score: AI vs Non-AI Users (Column Chart)

Compares average final scores between the two groups — both land around 56.8, showing that simply using AI tools does not, by itself, move the average score.

3

Correlation with Final Score (Horizontal Bar Chart)

Ranks factors by their correlation strength with final score. Last Exam Score and Assignment Avg show the strongest positive correlation, while AI Dependency and AI-Generated Content % show weak or negative correlation.

4

More AI Content = Lower Final Score (Bar + Trend Line)

Buckets students by the % of their work that is AI-generated (0-24%, 25-49%, 50-74%, 75-100%) and plots average score. The downward trend line shows scores fall as AI-generated content share rises.

5

AI Dependency Level vs Final Score (Column Chart)

Groups students into low (1-3), medium (4-6), and high (7-10) AI dependency bands. Scores stay flat at low-to-medium dependency but tick up slightly at high dependency, suggesting dependency alone isn't the deciding factor.

Performance — Academic Outcomes

Pass rates, score distribution, and how AI tool/purpose choice relates to final scores.

Overall Pass Rate

88.93%

High Performance Rate

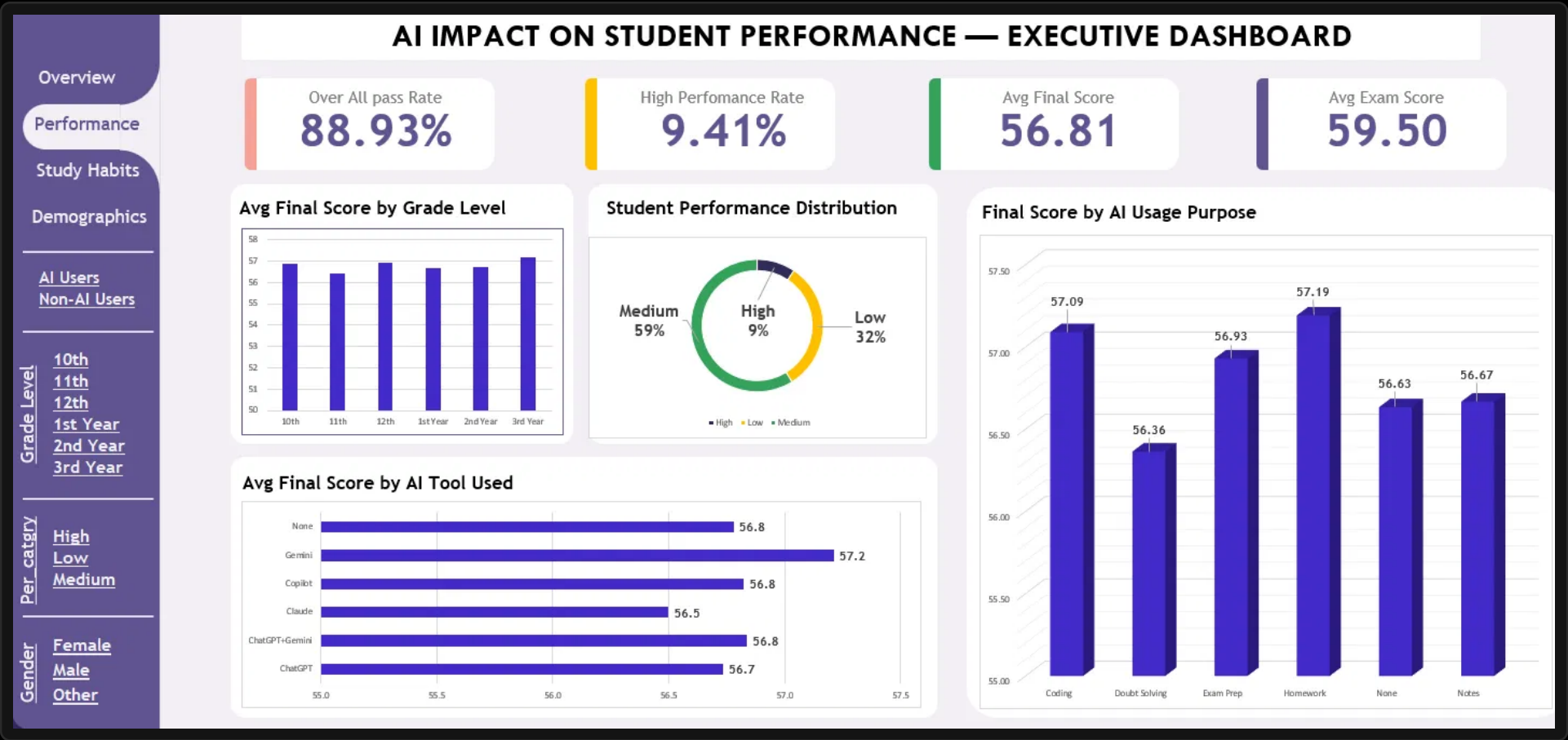
9.41%

Avg Final Score

56.81

Avg Exam Score

59.50



Performance — Chart Explanations

Chart-by-chart breakdown of what each visual shows and why it matters.

1

Avg Final Score by Grade Level (Column Chart)

Tracks average final score across 10th, 11th, 12th, and 1st-3rd year students. Scores stay in a tight 55-57 band, with 3rd Year students scoring highest overall.

2

Student Performance Distribution (Donut Chart)

Breaks the student body into High (9%), Medium (59%), and Low (32%) performers — showing the majority sit in the middle performance band.

3

Final Score by AI Usage Purpose (3D Column Chart)

Compares average scores by why students use AI — Homework (57.19) and Coding (57.09) score highest, while Doubt Solving (56.36) scores lowest, hinting that passive help lifts scores less than active-use cases.

4

Avg Final Score by AI Tool Used (Horizontal Bar Chart)

Ranks tools by average score of their users: Gemini (57.2) leads, followed by ChatGPT, Copilot and ChatGPT+Gemini (~56.7-56.8), with Claude lowest (56.5) — differences are small across all tools.

Study Habits — Behavior & Score

Attendance, sleep, social media, and study-hour patterns plotted against final score.

Study Efficiency Index

17.28

Avg Study Hrs

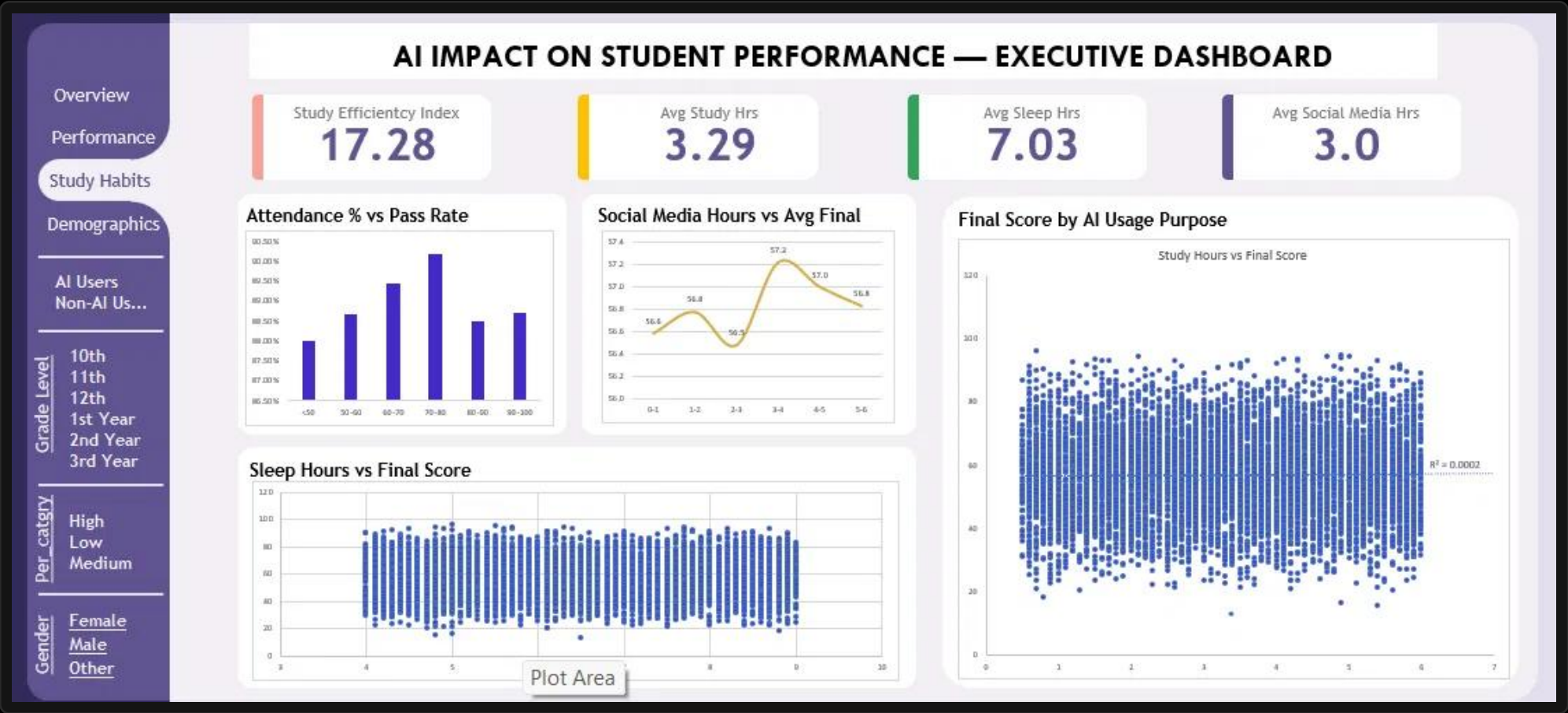
3.29

Avg Sleep Hrs

7.03

Avg Social Media Hrs

3.0



Study Habits — Chart Explanations

Chart-by-chart breakdown of what each visual shows and why it matters.

1

Attendance % vs Pass Rate (Column Chart)

Groups students by attendance bracket and shows pass rate per group. Pass rate rises with attendance up to the 70-80% band before leveling off, confirming attendance supports better outcomes.

2

Social Media Hours vs Avg Final (Line Chart)

Plots average final score against daily social media hours. Scores dip around 2-3 hours and recover afterward, showing no strong, consistent penalty from social media use alone.

3

Study Hours vs Final Score (Scatter Plot)

Plots every student's study hours against their final score. The near-flat trend line ($R^2 = 0.0002$) shows study hours alone barely explain score variation in this dataset.

4

Sleep Hours vs Final Score (Scatter Plot)

Plots sleep hours against final score for all students. Scores are spread broadly across 4-9 sleep hours, indicating no single sleep pattern strongly predicts performance.

Demographics — Gender & Grade View

AI adoption, scores, and participation broken down by gender and grade level.

Male Students

48.9%

Female Students

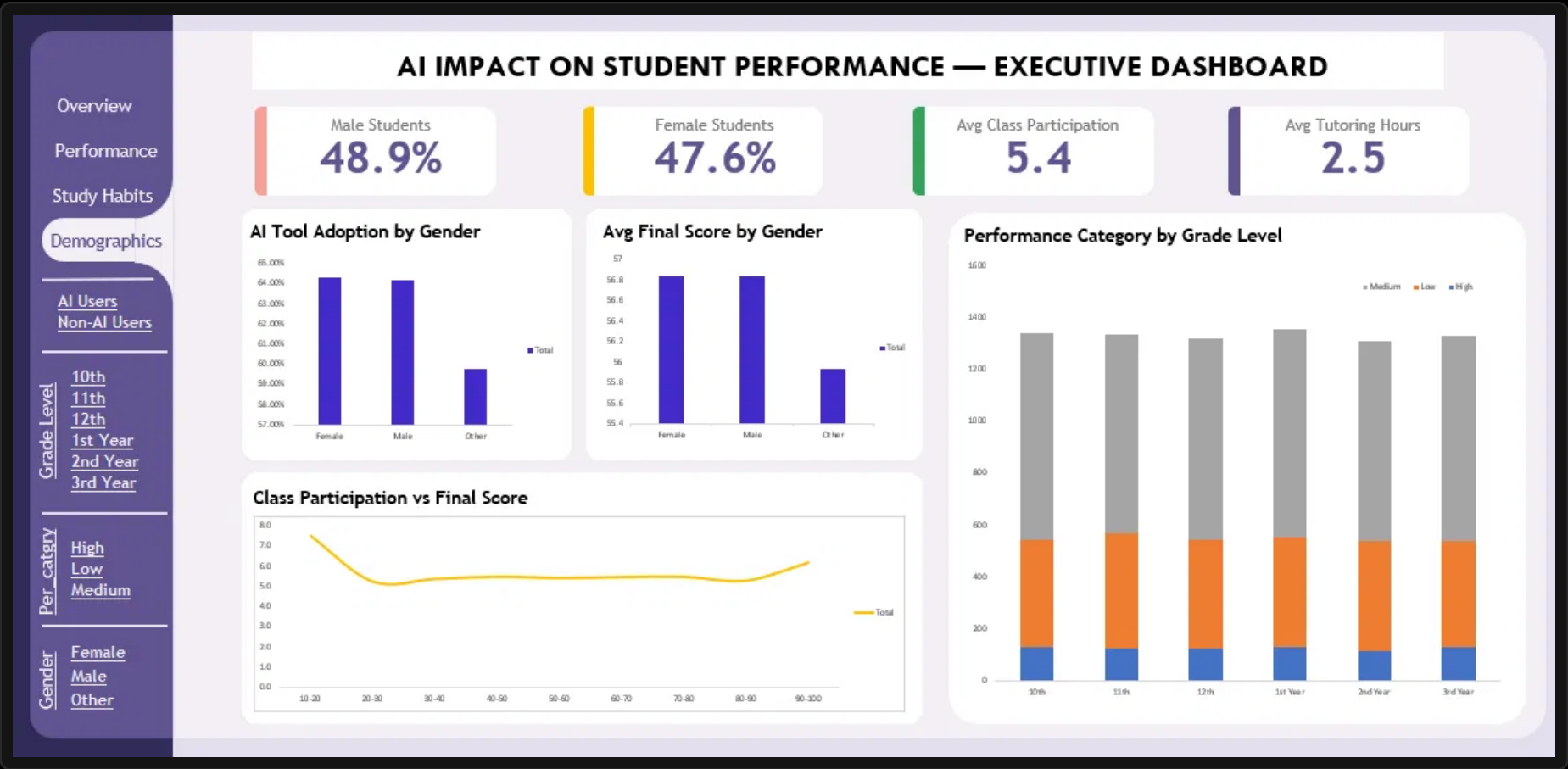
47.6%

Avg Class Participation

5.4

Avg Tutoring Hours

2.5



Demographics — Chart Explanations

Chart-by-chart breakdown of what each visual shows and why it matters.

1

AI Tool Adoption by Gender (Column Chart)

Compares AI adoption rate across Female, Male, and Other students. Female and Male students adopt AI at similar rates (~64%), while the Other category shows notably lower adoption.

2

Avg Final Score by Gender (Column Chart)

Shows average final score is nearly identical for Female and Male students (~56.7-56.8), with the Other group scoring slightly lower.

3

Class Participation vs Final Score (Line Chart)

Plots average final score against class participation score. Final score dips slightly at low-to-mid participation and recovers at the highest participation levels.

4

Performance Category by Grade Level (Stacked Column Chart)

Stacks High, Medium, and Low performers within each grade level. The Medium band dominates every grade consistently, showing performance mix is stable across the academic journey.

What the Data Tells Us

- 1 AI adoption is mainstream (64.1%) but using AI tools alone does not raise or lower average final scores in a meaningful way.
- 2 Heavier reliance on AI-generated content is linked to lower final scores — the more of a student's work AI produces, the more scores decline.
- 3 Traditional fundamentals — exam performance and assignment average — remain the strongest predictors of final score, far more than AI usage or dependency.
- 4 Attendance and class participation correlate more clearly with pass rates than study hours, sleep, or social media use do.
- 5 Gender and grade level show minimal impact on both AI adoption and final scores — outcomes are fairly consistent across groups.
- 6 Overall: AI is a widely-used study aid, but disciplined study habits and genuine understanding still drive academic performance.

Thank You